AI ENGINEERING / LOGISTICS / FINANCE OPERATIONS / CASE STUDY

FREIGHT INVOICE AUDIT

CASE STUDY

08

A synthetic distributor tests whether every carrier invoice can be rebuilt from contract terms.

PROPOSED RESEARCH / NOT YET VERIFIED

THE BUSINESS CONTEXT

A synthetic mid sized distributor ships thousands of parcels and pallets a month with several carriers. Each carrier bills with its own rate card and fuel table plus accessorials. Finance can only sample invoices so overcharges that follow the contract on paper go unseen.

WHO IS INVOLVED AND WHAT THEY NEED

01	**Finance controller**	Wants every dispute backed by exact amounts and evidence a carrier cannot argue with.
02	**Logistics manager**	Needs delivery facts such as waits and failed attempts to be right before anyone disputes.
03	**Procurement lead**	Owns contract terms and needs ambiguous clauses surfaced rather than silently interpreted.

SCOPE AND HONEST LIMITS

- ⚠ All carriers and contracts are synthetic so tariff complexity is likely understated.
- ⚠ No savings or recovery figures are claimed; only measurement on synthetic batches is planned.
- ⚠ Carrier responses to disputes cannot be simulated reliably.
- ⚠ Delivery event quality in a real business may be far worse than the synthetic feed.

CURRENT WORKFLOW AND FAILURE POINTS

08

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HOW THE WORK RUNS TODAY

1 TODAY

Invoices arrive

Carriers email PDFs and spreadsheets in different layouts at different times of the month.

2 TODAY

Manual keying

An analyst copies totals into a payment sheet and checks only the headline amount.

3 TODAY

Sample check

A few invoices a month are compared by hand against a rate card held in a shared folder.

4 TODAY

Fuel and extras

Fuel surcharge percentages and accessorials are accepted as billed unless something looks odd.

5 TODAY

Query by email

Suspected errors are queried informally with the carrier and the outcome is rarely logged.

6 TODAY

Pay and file

The invoice is approved for payment and filed with no record of what was checked.

WHERE IT BREAKS

Sampling blind spots	Most lines are never recalculated so small repeated overcharges pass unnoticed.
Scattered terms	Contracts and amendments sit in documents while rates live in separate spreadsheets.
Weak disputes	Claims without delivery evidence or exact amounts are easy for a carrier to reject.
No audit trail	There is no record of which checks ran or why a charge was accepted.

REDESIGNED WORKFLOW AND DECISION AUTHORITY

08

PROPOSED RESEARCH / NOT YET VERIFIED

THE REDESIGNED WORKFLOW

1 SYSTEM

Ingest and normalise

Invoices in each carrier layout are parsed into typed lines and linked to shipments.

→

2 SYSTEM

Rate from contract

Each line is rebuilt using the rate card version and fuel table in force on the ship date.

→

3 SYSTEM

Check delivery events

Accessorials such as waiting time and redelivery are tested against recorded delivery events.

4 SYSTEM

Variance and evidence

Differences are listed per line with the rule and inputs and expected against billed amount.

→

5 PERSON

Finance review

A finance person reads each evidence pack and chooses which variances to dispute.

→

6 PERSON

Dispute and learn

Raised disputes and carrier replies are logged and ambiguous clauses go to procurement.

SYSTEM STEP PERSON DECISION

WHERE AUTHORITY STAYS WITH A PERSON

Whether to raise a dispute	FINANCE CONTROLLER	A dispute affects the carrier relationship and only a person can weigh that cost.
How to read an ambiguous clause	PROCUREMENT LEAD	The commercial intent behind a contract term cannot be inferred by the system.
Whether to pay a disputed invoice	FINANCE CONTROLLER	Payment timing carries cash and service risk that sits with the business.

ARCHITECTURE, DATA AND CONTROLS

08

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SYSTEM ARCHITECTURE

1	**Document intake**	Parsers turn carrier PDFs and spreadsheets into typed invoice lines with source references.
2	**Contract store**	Versioned rate cards and fuel tables and accessorial rules with effective dates.
3	**Rating engine**	Deterministic recalculation of every charge line from shipment facts and contract rules.
4	**Evidence and variance**	Builds per line evidence packs showing rule and inputs and expected against billed amounts.
5	**Review workspace**	Queue for finance decisions with an immutable log of every choice and note.

CORE DATA ENTITIES

- **Carrier**
- **Contract version**
- **Rate card**
- **Shipment**
- **Delivery event**
- **Invoice line**

RISKS AND THE CONTROL FOR EACH

Wrong rate version applied →	Rate lookup uses the ship date and records the contract version on every recalculated line.
Model misreads an invoice →	Parsed totals must reconcile to the invoice header or the invoice goes to a person.
Disputing a valid charge →	Variances caused by missing amendments or unclear clauses are flagged and not recommended.
Silent system action →	The system cannot contact a carrier or change payment status without a person.

EVALUATION AND DELIVERY PLAN

08

PROPOSED RESEARCH / NOT YET VERIFIED

HOW SUCCESS IS MEASURED

Variance recall	Share of injected synthetic overcharges that the engine reports.
Amount accuracy	Gap between reported and injected overcharge per line and per invoice.
False dispute rate	Share of recommended variances a reviewer rejects as valid charges.
Unexplained variance	Value of differences the engine cannot attribute to a named rule.
Reviewer time	Minutes a person needs per invoice with and without evidence packs.

TEST PLAN

1. **Seed invoices with known overcharges and confirm each is found at the exact amount.**
2. **Change a rate card effective date and confirm older shipments still use the earlier rates.**
3. **Remove a delivery event and confirm the related accessorial is flagged as unverifiable.**
4. **Replay the same batch twice and confirm identical variance output and evidence packs.**

DELIVERY PLAN

01 ALIGN	02 UNDERSTAND	03 DESIGN	04 DELIVER	05 LAUNCH
Agree charge types and which decisions stay with finance.	Map rate cards and fuel tables and delivery data sources.	Design typed schemas and the rating rules and evidence format.	Build the engine and review workspace on synthetic data.	Run a controlled parallel pilot beside the current sampling.
DECISION Which charge types are in scope first	**DECISION** Which contract terms are authoritative	**DECISION** How ambiguity is surfaced to a person	**DECISION** Whether recall and false dispute rates are acceptable	**DECISION** Whether to extend to more carriers

DELIVERABLES

- Synthetic carrier contract and invoice dataset
- Rating engine specification and rule catalogue
- Variance evidence pack template
- Reviewer decision log design
- Evaluation report with failure cases